

Question				Marking details	Marks available														
					AO1	AO2	AO3	Total	Maths	Prac									
3.	(a)			Carbon-12, 6 neutrons and 6 protons or same number of p and n (1) Carbon-14, has 8 neutrons and 6 protons or more neutrons than protons (1) [So] C-14 is unbalanced or C-12 is balanced (1) Don't accept stable or unstable. N.B. treat as neutral any reference to electrons	3			3											
	(b)			<table><tr><td>gamma</td><td>γ (1)</td><td></td></tr><tr><td>alpha</td><td></td><td><u>helium nucleus</u> (1) Neutral: 2 protons and 2 neutrons</td></tr><tr><td>beta</td><td>${}^0_{-1}\beta$ or ${}^0_{-1}e$ (1)</td><td>[high energy / fast moving] <u>electron</u> (1)</td></tr></table>	gamma	γ (1)		alpha		<u>helium nucleus</u> (1) Neutral: 2 protons and 2 neutrons	beta	${}^0_{-1}\beta$ or ${}^0_{-1}e$ (1)	[high energy / fast moving] <u>electron</u> (1)	4			4		
gamma	γ (1)																		
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				Question 3 total	7	0	0	7	0	0									